

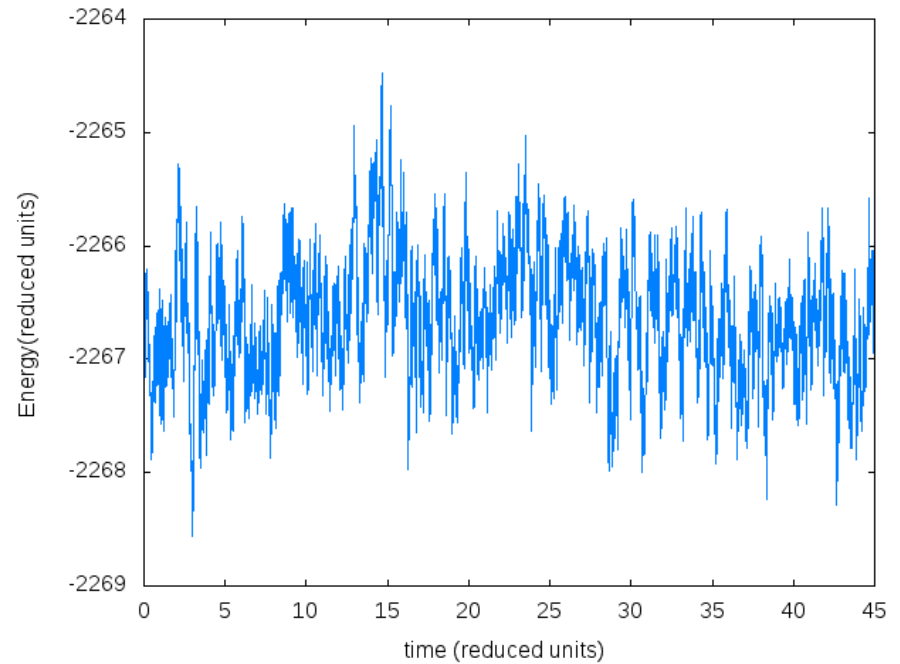
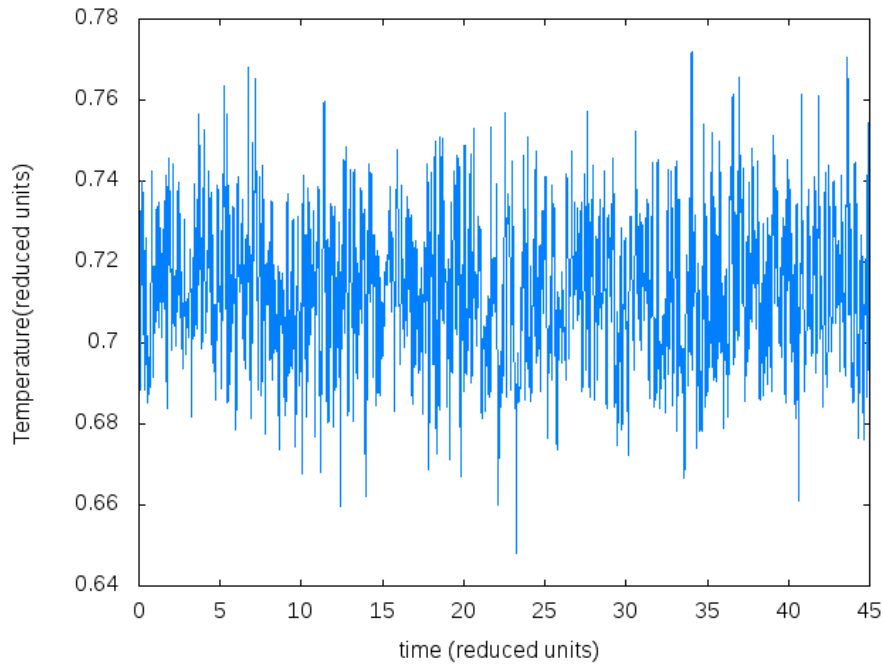
Practice 2. Molecular dynamics simulation of a Lennard-Jones system

Simulation of a simple Lennard-Jones fluid, such as liquid argon, with a hand-made FORTRAN MD code which incorporates the Leap-frog Verlet algorithm

STEP 1: Understanding the code.

- Open the FORTRAN file '**leapfroglj.f**' and have a look.
- Note that you'll need 2 auxiliary files:
(1) '**leap-lj.data**' and (2) '**leap-conf.data**', containing (1) data input and (2) initial coordinates and velocities, respectively.
- The main program is ready to simulate a LJ liquid of 500 atoms of mass 39.948 uma, $\sigma = 3.405 \text{ \AA}$ and $\epsilon/k_B = 119.8 \text{ K}$ (Argon), with a time step of $\Delta t = 0.01 \text{ ps}$. Remember that the LJ potential is of the form:
$$V_{LJ} = 4 \epsilon [(\sigma/r)^{12} - (\sigma/r)^6].$$

- Compile the code and make a run of 10000 time steps. If you compute the temperature T and total energy E (kinetic + potential) you will get the following figures:



STEP 2: Analysing the structure and calculating thermodynamic properties.

- Modify the code to calculate the **radial distribution function (RDF) $g(r)$** from the MD simulation. Compute the RDF for a run of 10000 steps. Plot $g(r)$ vs. r using reduced units.
- Modify the code to calculate the **pressure P** . Compute the mean values of the total and different contributions to E and P . **Include the tail corrections**. Analyze the convergence of the results as a function of the simulation length. Use reduced units.

STEP 3: Analysing the dynamics.

- Modify the code to calculate the **velocity autocorrelation function (VACF)** and the **mean square displacement (MSD)**:

$$C(t) = \langle \vec{v}(t) \cdot \vec{v}(0) \rangle$$

$$r^2(t) = \langle [\vec{r}(t) - \vec{r}(0)]^2 \rangle$$

Compute the VACF and the MSD for a run of 10000 steps. Plot the results.

- Write a code to calculate the **spectral density (SDW)** of the VACF:

$$S(\omega) = \int_0^\infty C(t) \cos(\omega t) dt$$

Use a Simpson 3/8 rule or similar. Plot the results.

- Use Einstein's diffusion formula to obtain the **self-diffusion coefficient D** :

$$r^2(t) = 6Dt$$